SPE-227455-MS

Leveraging Advanced Cuttings Analysis Towards Chemosteering Across Sands Productive Zones in Underbalanced Wells

Khalid Qubaisi and Azizi Ibrahim, Saudi Aramco; Haitham Ahmed and Caroline Magnier, Excellence Logging

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Underbalanced Coiled Tubing Drilling minimizes formation damage and lost circulation but poses challenges in horizontal wells with small diameters. This study highlights the advantages of rock cuttings analysis during UBCTD in clastic formations. Using XRF and XRD techniques, high-productivity zones were identified following elemental ratios and mineral analysis in quartz-dominated lithology. The result of study demonstrated that this workflow has potential to replace LWD technologies, enhancing precision in chemosteering and well placement in complex formations.

The method combines microscopic observations to assess grain sorting and arrangement with systematic quantitative mineral analysis using X-Ray Diffraction (XRD) and X-Ray Fluorescence (XRF) techniques for detailed elemental composition of cuttings from eight lateral sections from different wells. Delimited zones are tentatively correlated with hydrocarbon productivity, offering valuable insights into clastic formations characteristics and their relationship to well placement and productivity.

Clay minerals and anhydrite cement significantly influence the reservoir quality of the formation, particularly by reducing porosity and permeability. These effects are primarily dictated by the depositional environment and subsequent diagenetic processes. In clastic formations the heterogeneity of depositional settings results in the development of distinct facies, ranging from eolian dune systems to interdune and sand sheet deposits, each exhibiting unique porosity and permeability characteristics. Accurate identification and differentiation of these facies in real-time is critical for chemosteering operations and optimizing hydrocarbon recovery.

While the XRD confirmed the purity of various types of clastic reservoir minor changes in clay content (kaolinite, illite) and presence of feldspars in traces identified depositional variations suggested to be related to eolian vs glaciation sedimentations. Similarly, elemental data showing low content of calcium, iron, magnesium, aluminum and potassium oxides with presence of strontium, as well as a key trace elements Zn, Zr, Ni and Rb to name just a few, allowed for better characterization of sandstones purity.

Characterizing rocks through microscopic analysis and quantitative compositional evaluation can be highly effective for navigating complex well trajectories within slim hole sizes, where downhole tool deployment is challenging. This approach demonstrates significant potential in enhancing reservoir

understanding and performance in clastic formations ensuring accurate facies identification and enhanced well placement.

Introduction

The studied geological sequence represents a significant clastic deposit in the subsurface of the Arabian Peninsula. It comprises mainly low-porosity sandstones, siltstones, and shales, deposited in glacial, fluvial, and aeolian settings. These depositional systems reflect a dynamic paleoclimatic regime transitioning from cold glacial conditions to more temperate fluvial and arid aeolian regimes (Al-Husseini, 1997). The mineralogical assemblage is dominated by quartz, feldspar, and clay minerals such as illite and kaolinite, pointing to a high siliciclastic input and variable depositional energy regimes (Alsharhan & Nairn, 1997; Al-Ramadan et al., 2012).

The geological interval is informally subdivided into three Units (Figure 1), each reflecting distinct lithological characteristics and reservoir quality. Unit 1, the uppermost interval, consists predominantly of fine- to medium-grained, well-sorted sandstones interbedded with siltstones and shales. Unit 2 is more heterogeneous, comprising interbedded sandstone, conglomerate, and mudstone, typically with higher clay content and poorer sorting. Unit 3, the lowermost Unit, is composed of medium- to coarse-grained quartz-rich sandstones with minimal clay, making it the most texturally and mineralogically mature of the three.

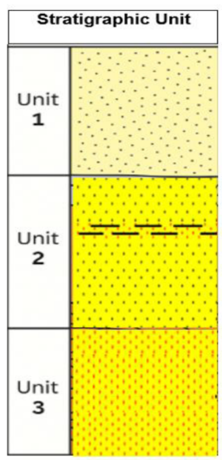

Figure 1—Legend for stratigraphic Units defined in this study. Color coding is illustrative only and not to scale

Although Unit 2 was identified in the stratigraphic framework, this study focuses solely on Unit 1 and Unit 3 due to their greater reservoir relevance.

These formations offer significant hydrocarbon potential, particularly in its cleaner sandstone intervals. Several operational constraints challenge the evaluation of reservoir quality in underbalanced wells. Slim hole sizes, complex borehole trajectories and instability of the borehole wall elevate the risk of tool loss or mechanical sticking, thereby limiting the deployment of conventional downhole wireline logging tools. In addition, high bottom hole temperatures often exceed the functional tolerance of many standard tools, further complicating in-situ logging operations.

Given these limitations, surface-based analytical techniques have become increasingly valuable in reservoir characterization workflows. These methods including X-ray fluorescence (XRF) elemental analysis, X-ray diffraction (XRD) mineralogical analysis and digital microscopy are non-intrusive and rely on the analysis of drill cuttings retrieved at the surface. Such technologies enable real time elemental and mineralogical analysis that can substitute or supplement downhole measurements, particularly in challenging well conditions.

The present study aims to characterize the mineralogical and elemental composition of the formation using surface based analytical techniques in order to evaluate its reservoir quality. Specific objectives include assessing the cleanliness of sandstone intervals as an indirect proxy for optimal hydrocarbon flux. Moreover, the study aims to verify well placement relative to the reservoir zones through geo-monitoring, to define reservoir thickness and net pay based on mineralogical markers, and to evaluate clay content (e.g., illite and kaolinite) as a determinant of reservoir quality. Ultimately, the results inform completion strategies and stage simulation design in high-productivity intervals, enhancing hydrocarbon recovery efficiency under complex drilling conditions.

Methodology and Analytical Approach

This study implements a detailed workflow for analyzing cutting samples collected across lateral sections of various wells. The primary focus is on geochemical and mineralogical characterization using X-ray fluorescence (XRF), X-ray diffraction (XRD), and digital microscopic evaluation, including grain sorting assessments. The methodology emphasizes standardized sample preparation and consistent analytical procedures to ensure accuracy and inter-sample comparability (Potts & Webb, 1992; Jenkins & Snyder, 1996).

Sample Acquisition and Preparation

Cutting samples were collected directly from the sand accumulator during drilling operations at one-hour intervals (The sand accumulator used during underbalanced coiled tubing drilling (UBCTD) isolates and captures rock cuttings from the returning drilling fluid through a cylindrical metallic filter basket (Figure 2). Following retrieval, the samples were subjected to a controlled cleaning process to eliminate contaminants and were subsequently air-dried using a vacuum system. Each sample was visually examined using a high-definition digital microscope and photographed for archival purposes. Descriptions were recorded for each sample, including observations on color, grain size, sorting and texture (Ulrych et al., 2018).

Figure 2—Sand accumulator used during UBCTD operations, showing external chamber and internal metallic mesh used to capture cutting materials

Cleaned samples were then crushed to a fine powder (<150 μm) using a mixer mill to ensure homogenization prior to geochemical and mineralogical analysis. For XRF analysis, approximately 5 grams of powdered sample were pressed into pellets using a 15-ton hydraulic press. For XRD analysis, about 0.5 grams of powder were prepared to enable consistent detection of crystalline phases. All analytical procedures were conducted following standardized protocol to guarantee data integrity and reproducibility (Qubaisi, Haitham, Haceb & Magnier, 2023).

Microscopic Examination

High resolution digital petrographic microscopy was utilized to characterize the physical features of the cuttings (Figure 3). This visual inspection assisted in identifying lithological changes and supported integration with the geochemical and mineralogical datasets. Archived images of the samples served as references to help correlate lithological variations with XRF and XRD results (Kh. Qubaisi, Haitham, Haceb & Magnier, 2023).

Figure 3—Example of cuttings sample under digital microscope, revealing grain angularity, sorting, and color variations

X-ray Fluorescence (XRF) Analysis

XRF analysis was utilized to determine the elemental composition of the cutting samples, focusing on both major and trace elements. The technique relies on bombarding the sample with primary X-rays generated by either a palladium (Pd) or tungsten (W) anode. When atoms are exposed to X-rays, inner-shell electrons are ejected and the subsequent transition of outer-shell electrons to fill these vacancies results in the emission of secondary X-rays with element-specific energies (Barkla 1909). This emitted energy is then detected and quantified (Figure 4).

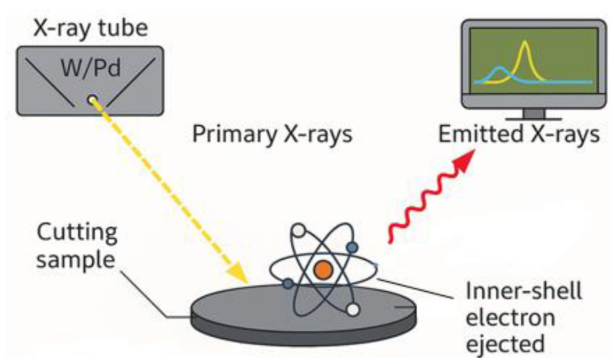

Figure 4—X-ray fluorescence (XRF) schematic illustrating interaction of primary X-rays with sample atoms and detection of element-specific emitted X-rays

Major elements were reported in weight percent (wt.%) while trace elements were reported in parts per million (ppm). To ensure the reliability of major and trace element readings, the instrument was regularly calibrated with certified geochemical reference materials (GRMs), and repeatability tests were conducted throughout the measurement sequence. This approach ensured that even trace-level variations remained analytically valid and interpretable within the stratigraphic framework.

Quality assurance included the analysis of geochemical reference materials with each batch of samples to monitor instrument stability and calibration. Instrumental drift and analytical consistency were evaluated and corrected where necessary. All datasets were reviewed by qualified data analysts for internal consistency and accuracy.

X-ray Diffraction (XRD) Analysis

XRD analysis was conducted to quantify the mineralogical composition by measuring the diffraction patterns produced when X-rays interact with the crystalline structure of the sample. Diffraction patterns are captured by a detector and are characteristic of each mineral. By analyzing the position and intensity of the diffracted peaks, both the atomic structure and the relative abundance of minerals in the sample can be determined. (Figure 5).

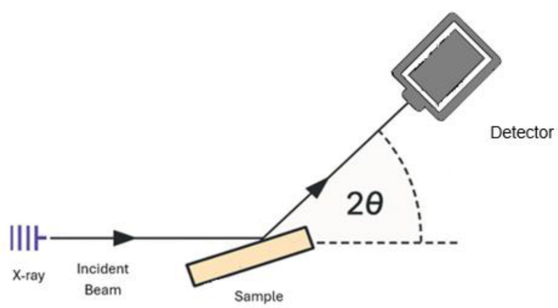

Figure 5—X-ray diffraction (XRD) setup showing the principle of Bragg's Law with incident beam, sample angle, and diffraction capture

Minerals identifiable but not necessarily present in the samples studied include quartz, K-feldspar, plagioclase, illite, kaolinite, illite-smectite, calcite, dolomite, siderite, pyrite, gypsum, Halite and anhydrite. The resulting mineralogical data were normalized to 100% of the detected crystalline phases (Figure 6).

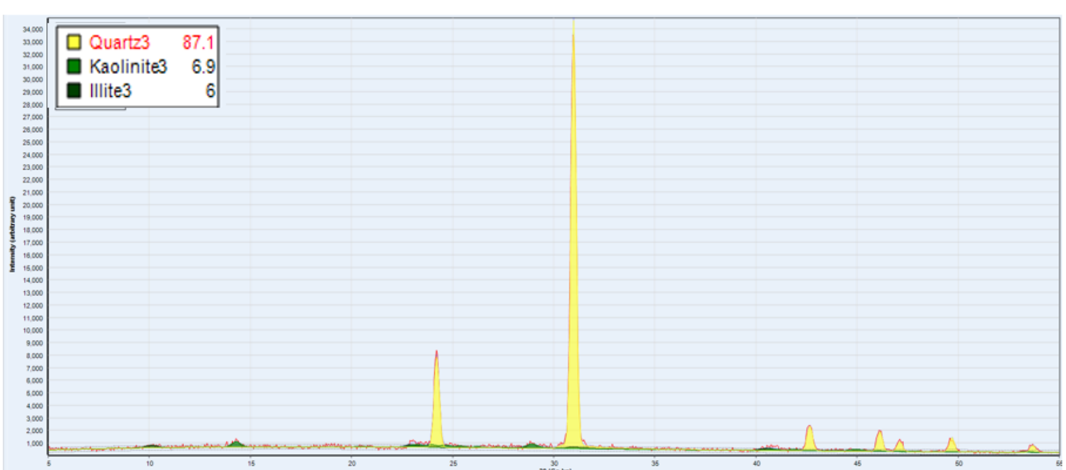

Figure 6—Example X-ray diffractogram from this study. This illustration is for reference only and does not represent a specific sample

Results and Discussions

Chemostratigraphic Trends Across Reservoir

The integration of XRD and XRF datasets from eight lateral wells enabled the establishment of a robust chemostratigraphic framework across the studied reservoir interval. Although the sequence is overall quartz dominated (>90%), consistent variations in elemental concentrations and clay mineralogy allowed for effective discrimination between the two primary target Units (Figure 7, Figure 8). The other detected minerals are clays: illite and kaolinite. Distinctions of the two chemostratigraphic units are clearly noted with most major elements, silica excluded as it is overabundant in the quartz main lithology. Trace elements were found to have valuable chemical information relating to sedimentary processes.

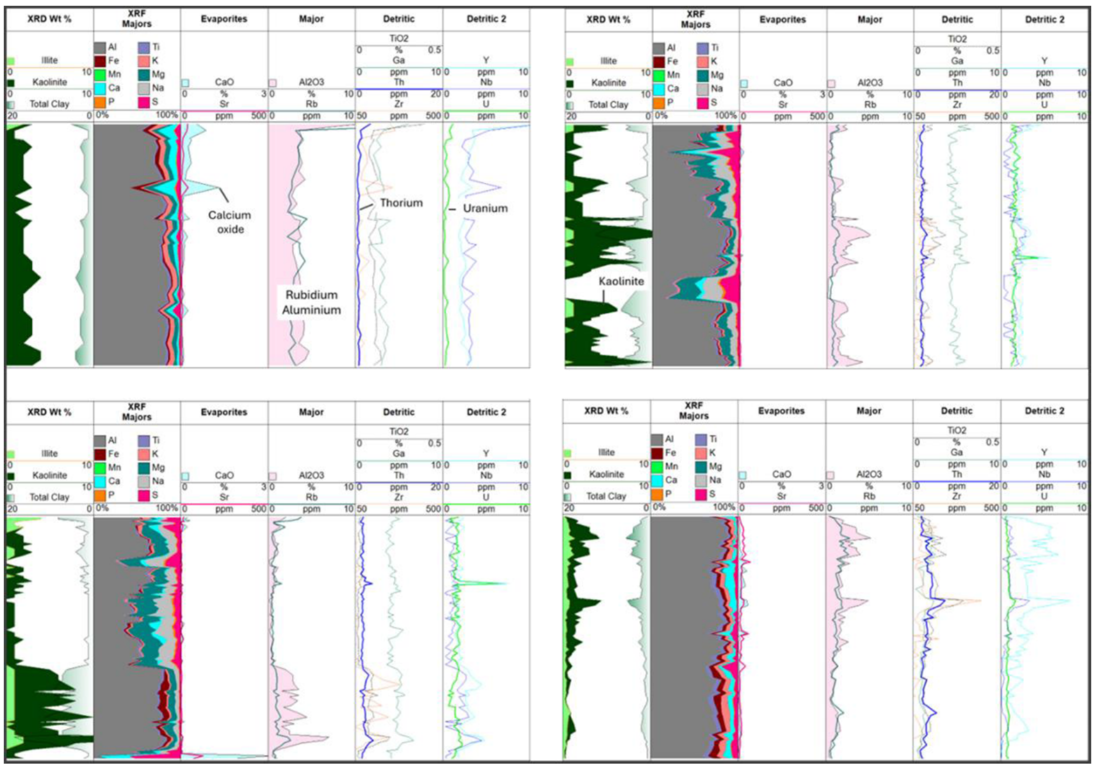

Figure 7—Unit 1 composite logs from 4 lateral wells highlighting Rb and Al_2O_3 detritic content low Th and CaO and low U. Note presence of kaolinite in clays.

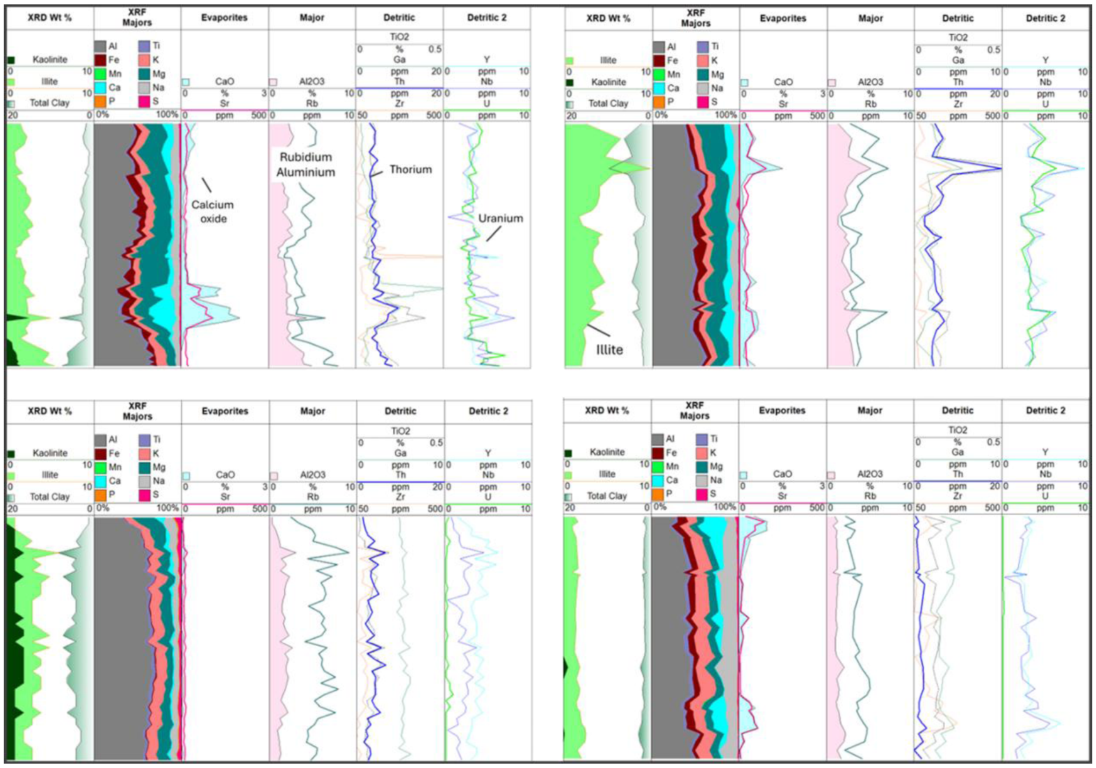

Figure 8—Unit 3 composite logs from 4 lateral wells showing elevated Th, Rb, and Al_2O_3, in illite-dominated sedimentary intervals. Note the absence of kaolinite in these stratigraphic Units.

The sandstone in Unit 1 is fine- to medium-grained and moderately sorted. It shows a higher content of kaolinite over illite indicating probable K-feldspar alterations (as noted also with lower K_2O). These elemental and mineral rock compositions and patterns suggest deposition in weathering context (Figure 7). Unit 1 exhibited moderate to high concentrations of Al_2O_3 (ranging from 2% to 5%) and lower relative content of Rb. The calcium oxide content increases at some depth intervals due to silicates, found together correlating with a minor strontium spikes, when clays are altogether less present, indicating likely some paleo incursions of saline or anhydrites that are not detected by mineral analysis.

Unit 3 in contrast, is defined by a significant increase in quartz purity (>98%) and a marked reduction in clay-related elements. The sandstones are medium- to coarse-grained, well sorted, and texturally mature, with minor illite (1–2%) and are quite depleted in kaolinite. The sedimentary features indicate a difference in deposition or processes in the two units. Unit 3 with higher quartz is also consistent with expected higher reservoir quality and would make Unit 3 the primary productive target. Although quartz-dominated sandstones appear visually clean, trace elements such as zirconium (Zr), rubidium (Rb), thorium (Th), and uranium (U) were consistently detected in low but geologically meaningful concentrations. These elements are typically hosted in accessory minerals such as zircon, monazite and apatite, or associated with clay and K-feldspar phases. Their persistence, even in texturally mature and mineralogically pure intervals, highlights subtle detrital or diagenetic contributions and provides valuable chemostratigraphic markers. These trace elements enable refined differentiation between otherwise similar-appearing sandstone intervals, enhancing resolution in depositional interpretation and well placement decisions.

The chemostratigraphic contrast between these two Units supports their differentiation in lateral sections and provides critical input for real-time geo-monitoring and well placement optimization.

Their consistent presence provides additional resolution for chemostratigraphic interpretation, supporting differentiation of subtle facies changes not evident from major elements alone.

The upper Unit (Unit 1) is consistent with literature publications and has been interpreted as eolian-dominated sandstones, exhibiting relatively homogeneous mineralogical profiles. XRD analysis consistently showed these Units to be composed predominantly of quartz and kaolinite.

While some Unit 3 intervals also displayed high kaolinite, most were characterized by a dominance of illite alongside quartz. These findings support a more variable and potentially heterogeneous detrital cycles altering depositional inputs in the lower Unit (Figure 8). In Unit 3 rock samples, although similar Al_2O_3 content, Rb is higher and more easily linked to detritic input from illite clays. Trace elements U, Y, and Nb are also in higher content. These elements corroborate more continental, fluvial and/or glacial or postglacial dynamic fluvial settings.

These trends were represented in major element cross-plots, particularly K_2O vs Al_2O_3, Fe_2O_3 vs TiO_2 and CaO vs MgO (Figure 9). Others like major elements S or NaO which add to the same obvious mineral variability where sulfur is found in higher concentrations in Unit 1 and NaO highest in Unit 3 rock samples. Likewise, in Unit 1 there are lower Fe_2O_3 and TiO_2 values exhibited as compared to Unit 3.

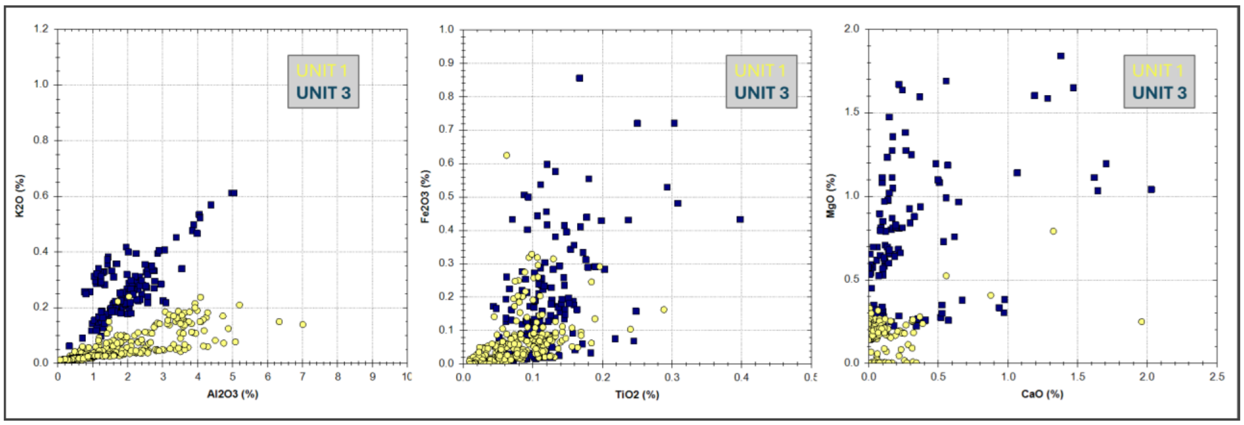

Figure 9—Cross-plots of major elements (Al_2O_3, K_2O, Fe_2O_3, TiO_2, CaO, MgO) clearly distinguishing Unit 1 (kaolinite-rich) from Unit 3 (illite-dominated)

Elemental Associations and Geochemical Proxies

The analysis of trace elements further supported mineralogical interpretation and facies differentiation. The depth logs of each well were shown on figures 7 and 8 and showed trends that were confirmed with cross-plots. For instance, uranium (U) vs thorium (Th), yttrium (Y) vs zirconium (Zr) also show systematic variability. Strontium (Sr) vs Rubidium (Rb) concentrations can relate to one Unit type, although some chemical heterogeneity is also evidenced due to the dataset. Most of the chemo-associations highlight the influence of clay content and redox conditions, with U being enriched in clay-rich intervals and Th and Zr more abundant in quartz-rich Units (Figure 10).

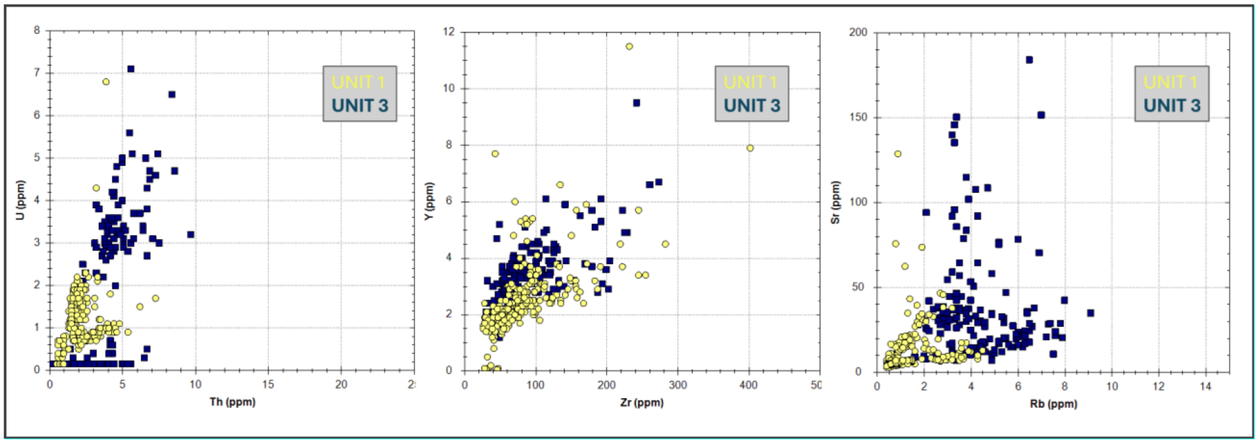

Figure 10—Cross-plots of trace elements (U vs Th, Y vs Zr, Rb vs Sr) identifying clay- and sand-dominated intervals and potential redox variation

Specific geochemical ratios used were of particular interest to infer depositional settings and mineralogical character. The Zr/TiO_2 ratio, often interpreted as a proxy for sedimentary heavy mineral concentration, indicated that the lower Unit sandstones more compact from sediment load, both texturally and compositionally, than their upper counterparts. Variations in Fe_2O_3 content were observed across both categories, although these were interpreted to reflect localized chemical alterations rather than differences in primary depositional input (Figure 9, Figure 11).

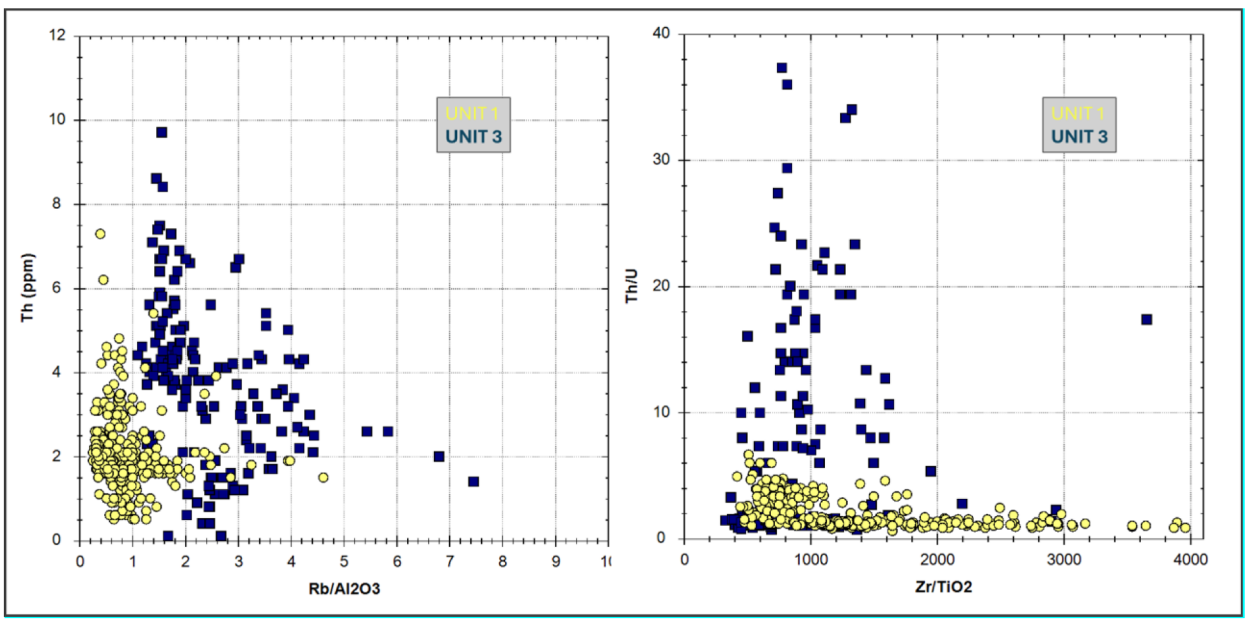

Figure 11—Rb/Al_2O_3 vs Th and Zr/TiO_2 vs Th/U ratios illustrating Sandstone maturity and redox variability across stratigraphic Units

Petrographic and Sorting Observations

Microscopic and petrographic analyses further validated the geochemical and mineralogical trends observed in the bulk analyses. Samples from the lower Unit revealed well-sorted quartz grains with minimal feldspar or lithic inclusions, confirming high mineralogical purity (>95%). In contrast, samples from the upper Unit showed more moderate sorting and a visibly higher abundance of clay minerals, often filling pore spaces and reducing grain contact (Figure 12). These textural differences are significant as they correlate and suggest differences in permeability between the two Units (Unit 1 and Unit 3).

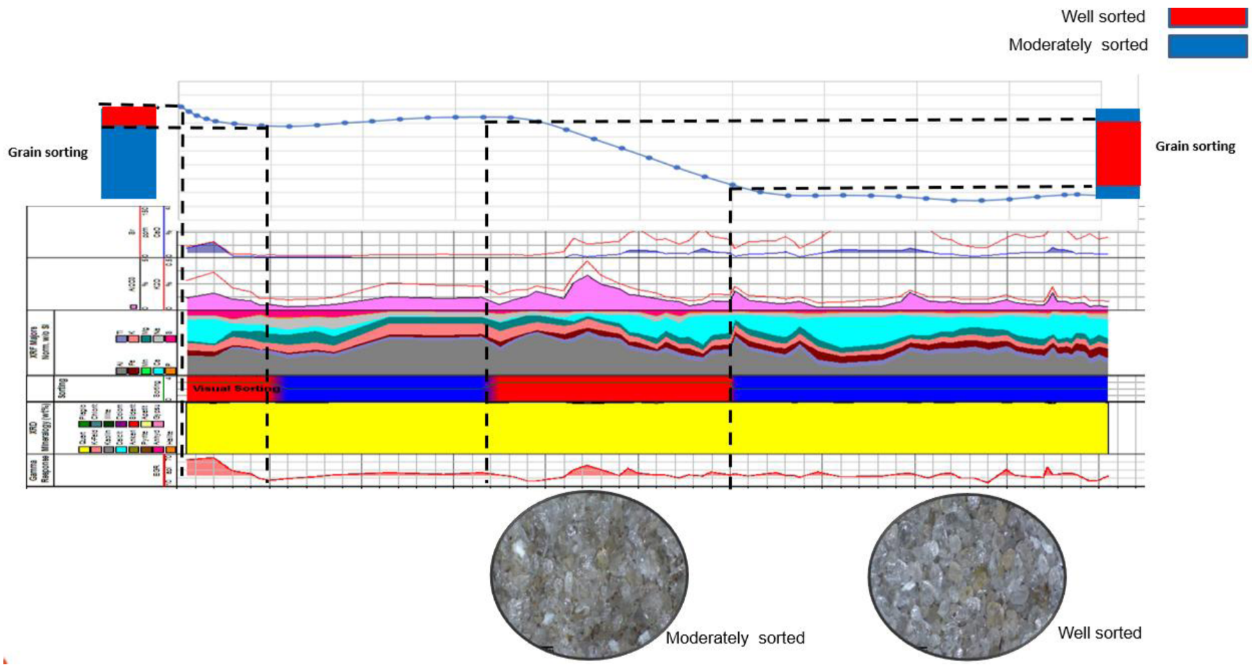

Figure 12—Integrated log panel and petrographic evaluation illustrating grain sorting and clay content across sandstone intervals. Well-sorted intervals dominated by quartz-rich grains are confirmed through visual inspection and digital microscopy. Moderately sorted zones show higher clay content, which likely reduces porosity and grain connectivity. The images below the log panel represent the contrasting textures between the two main Units

Case-Based Observations from Lateral Wells

Field-scale application of these findings across eight horizontal well, four intersecting upper Units and four intersecting lower Units reinforced the robustness of the geochemical framework. Trends observed in the content of major oxide elements K_2O and Al_2O_3 corresponded well with clay mineral concentrations, while sodium oxide (Na_2O) levels appeared to distinguish two lower Unit wells that show probable plagioclase content.

Trace element behavior also aligned with the proposed facies model. For example, iron vs titanium plots showed a roughly linear relationship, with the upper Unit (Unit 1) generally displaying lower values for both elements (Figure 9). Additionally, while many trace elements behaved consistently across wells, certain deviations were noted, suggesting either heterogeneity in sediment input or variable diagenetic alteration (Figure 9, Figure 10).

Summary of Key Geochemical Ratios

Several elemental ratios were particularly effective in discriminating between lithofacies and evaluating reservoir properties. The Rb/Al_2O_3 ratio differentiated illite-rich lower Units (Unit 3) from kaolinite-rich upper Units (Unit 1). The Zr/TiO_2 ratio is a reliable indicator of sedimentary maturity change through weathering process and sorting, while Th/U and Th/Ga ratios proved useful in tracking redox and clay variability across the stratigraphy Figure 11).

Geosteering Application

In multiple wells, this approach enabled the optimization of lateral well trajectories. For example, increasing illite and CaO concentrations were used to redirect drilling trajectories back into productive zones. In other instances, a marked rise in Al_2O_3 and clay-associated elements indicated reservoir exit, leading to timely total depth (TD) decisions. In thinly bedded intervals, continuous high quartz concentrations flanked by kaolinite-rich zones confirmed lateral continuity and allowed for confirmation of pay zones despite ambiguous conventional log data (Figure 13, Figure 14).

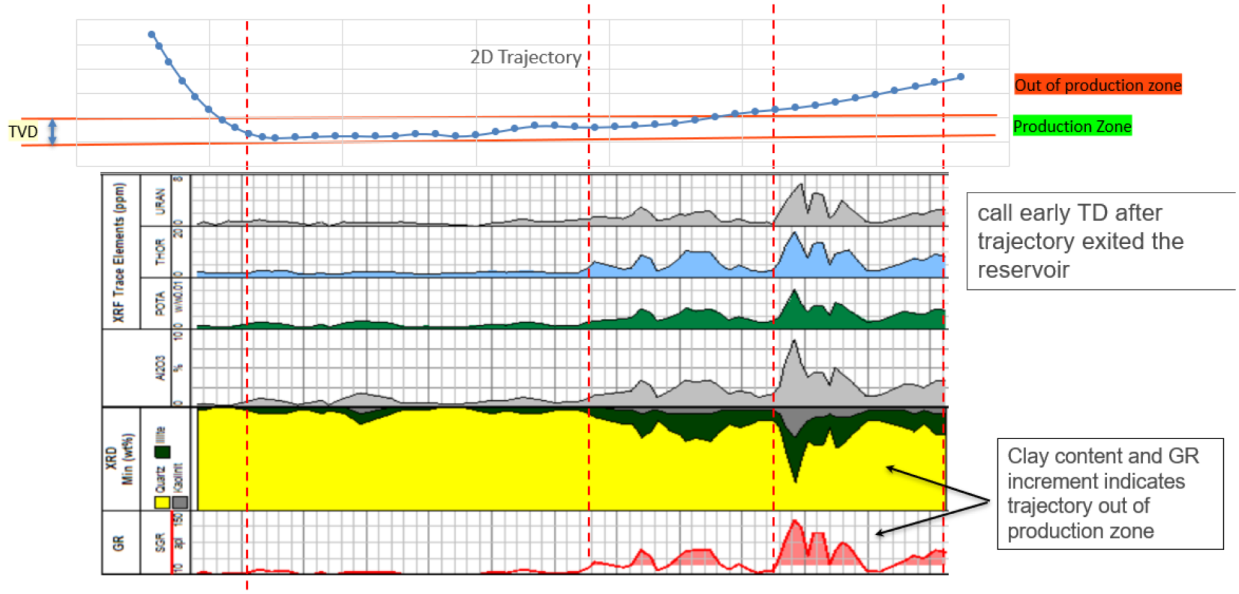

Figure 13—Real-time geomonitoring of trajectory through the productive zone. An early Total Depth (TD) decision was triggered after a sudden increase in clay content and gamma ray (GR) response. The well initially remains within the reservoir, indicated by stable geochemical signatures and low GR values. As drilling progresses, the trajectory deviates upward, exiting the productive interval. This is marked by elevated clay and GR readings, increased elemental concentrations (e.g., Al, K), and a reduction in quartz content as shown by XRD and XRF data. These shifts clearly define the reservoir exit and demonstrate how integrated mineralogical and elemental data can support timely geosteering decisions and optimize reservoir exposure.

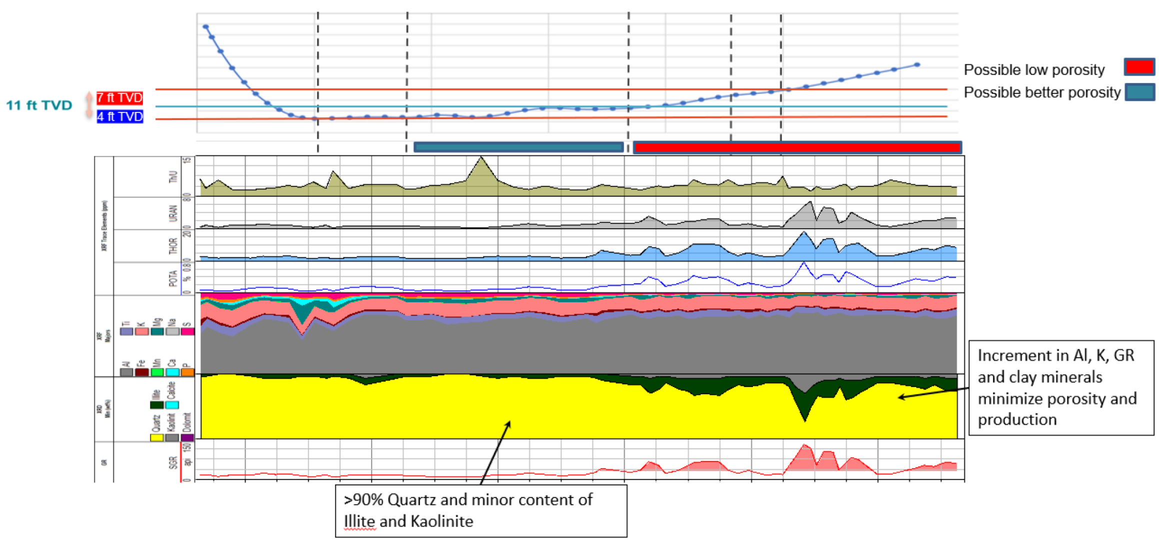

Figure 14—Mineralogical and elemental interpretation log showing porosity trends in relation to aluminum (Al), potassium (K), gamma ray (GR), and clay mineral content. Zones with increased Al, K, GR, and clay minerals (primarily illite and kaolinite) correspond to reduced porosity and are flagged as possible low-porosity intervals. In contrast, intervals dominated by >90% quartz with minimal clay content are associated with better porosity and reservoir quality

These findings support the broader adoption of chemostratigraphic tools for reservoir navigation and zonation, particularly in challenging drilling environments such as slim hole Under Balance Coil Tube Drilling (UBCTD) wells, where downhole logging is often limited or absent.

Conclusion

The results demonstrate that despite the overall quartz-rich nature of the studied sequence, subtle variations in mineralogy and geochemistry can effectively delineate internal stratigraphic categories and predict reservoir quality. The distinct geochemical signatures of the upper and lower Units reflect both their depositional histories and subsequent diagenetic overprints.

The upper Unit (Unit 1), with its higher kaolinite content, suggests deposition with significant K-feldspar diagenetic impact and weathering characteristic of Eolian deposits for example such as those found in subaerial weathering-dominated conditions. This is visible with the major element K_2O content depletion in Unit 1. The latter also has shown to have poorer grain sorting and expected reduced reservoir quality with lower porosity. In contrast, the higher K_2O, Rb, and Zr/TiO_2 values in the lower Unit (Unit 3) suggest a more continental input and confirmed with higher content of illite. The prevalence of illite further supports burial-related transformation processes, potentially linked to glacial or high-energy fluvial deposition.

These geochemical distinctions are significant because they are directly applicable to real-time drilling operations. The integration of GEAR-acquired XRF and XRD data in real-time allowed for chemosteering in environments where conventional logs (e.g., spectral gamma ray) were inconclusive due to borehole conditions. Shifts in elements such as Ca and Mg, as well as Al and Rb, served as reliable indicators for reservoir entry and exit (Figures 13 and 14).

Nomenclature

XRF	- X-Ray Fluorescence - Terminology
XRD	- X-Ray Diffraction - Terminology
UBCTD	- Underbalanced Coiled Tubing Drilling - Terminology

References

Al-Hajri, S. M., Hughes, G. W., & Al-Qahtani, M. (2009). Paleozoic stratigraphy and hydrocarbon habitat of the Arabian Plate. *GeoArabia*, **14**(3), 101–130.

Al-Husseini, M. I. (1997). Paleozoic stratigraphy of the Arabian Plate. *GeoArabia*, **2**(3), 361–384.

Al-Ramadan, K., Worden, R. H., & Satterley, A. K. (2012). Diagenesis and reservoir quality of the Permian Unayzah Sandstone, central Saudi Arabia. *Marine and Petroleum Geology*, **38**(1), 178–193.

Alsharhan, A. S., & Nairn, A. E. M. (1997). *Sedimentary Basins and Petroleum Geology of the Middle East*. Elsevier.

Alsharhan, A. S., & Whittle, G. L. (1995). Depositional environments of the Unayzah Formation in Saudi Arabia. *AAPG Bulletin*, **79**(1), 1–20.

Jenkins, R., & Snyder, R. L. (1996). *Introduction to X-ray Powder Diffractometry*. Wiley-Interscience.

Konert, G., Afifi, A. M., Al-Hajri, S. A., & Droste, H. J. (2001). Paleozoic stratigraphy and hydrocarbon habitat of the Arabian Plate. *GeoArabia*, **6**(3), 407–442.

Qubaisi, K., Haitham, A., Haceb, M., & Caroline, M. (2023). *Reservoir Characterization Utilizing Advanced Geochemical Measurements Acquired at Surface*.

Stephenson, M. H., Powell, J. H., & Al-Masri, A. (2003). A new model for the glaciogenic rocks of the Late Ordovician to Permian Haushi Group, Oman. *GeoArabia*, **8**(2), 251–272.

Ulrych, J., Hladil, J., Pruner, P., & Žák, K. (2018). Petrographic and geochemical techniques for cuttings analysis. *Geological Journal*, **53**(2), 304–318.